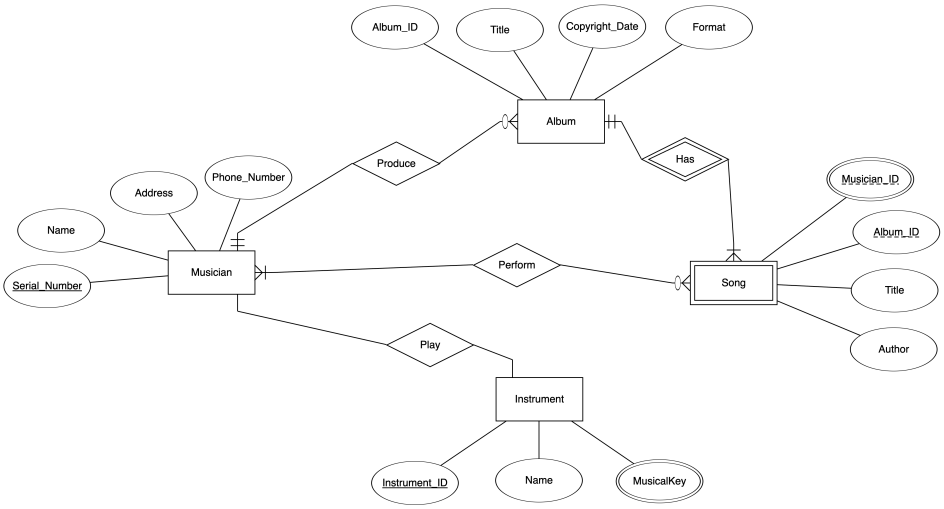
Assignment 2 - CCS 5001 Database

Question 1



Musician

Serial_Number	Name	Address	Phone_Number

Instrument

Instrument_ID	Name	MusicalKey

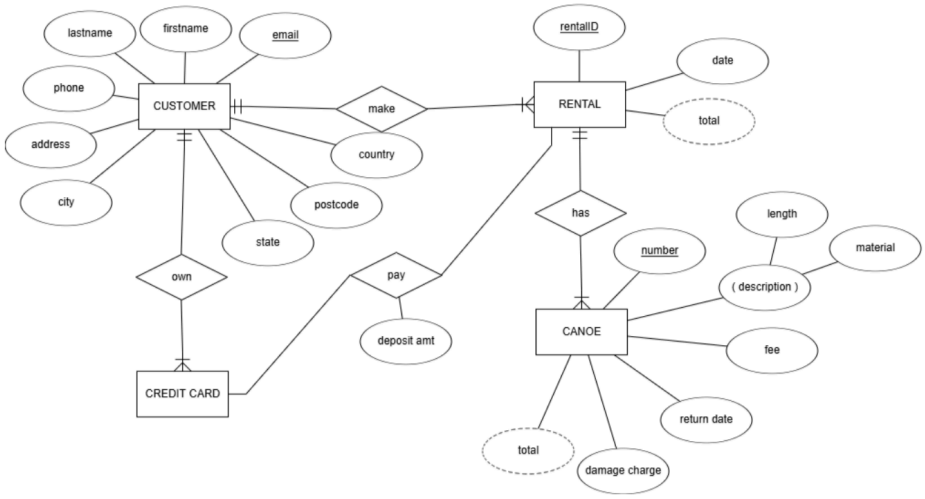
Song

Musician_ID (FK)	Album_ID (FK)	Title	Author

Album

Album_ID	Title	Copyright_Date	Format	Musician_ID (FK)

Question 2



Customer

Email	Firstname	Lastname	Phone	Address	City	State	Country

Credit Card

Email (FK)

Payment

Email (FK)	Deposit_Amount	rental_ID

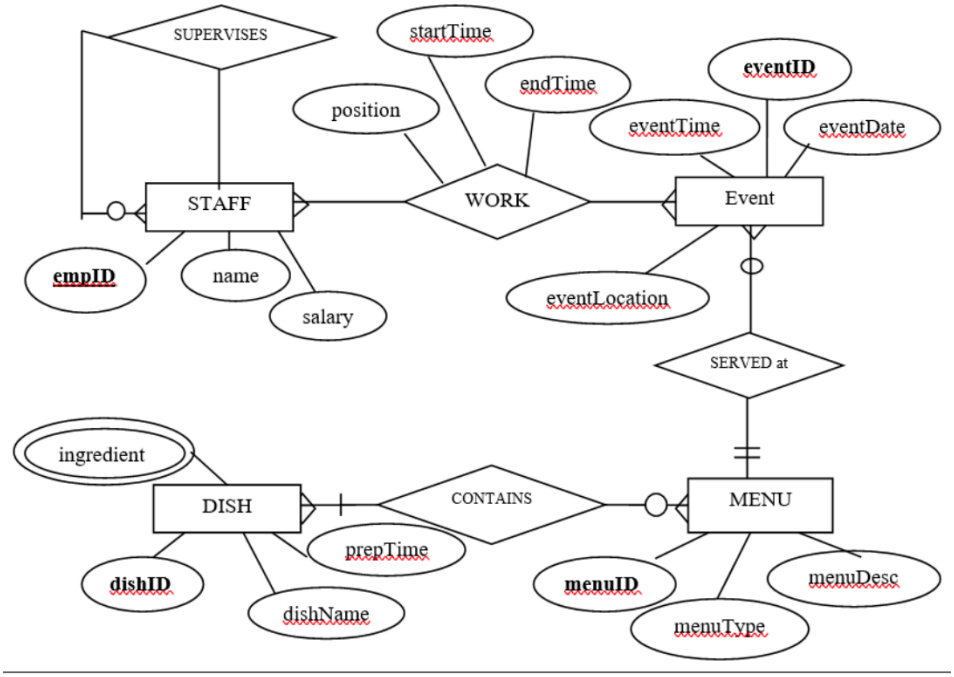
Rental

rental_ID	date	total	Email (FK)

Canoe

Number	length	material	fee	Return_Date	Damage_Charge	total	Rental_ID

Question 3



Staff

empID	Name	Salary	SupervisorID (FK)

Event

eventID	eventTime	eventDate	eventLocation	menuID (FK)

Work

eventID (FK)	empID (FK)	position	startTime	endTime

Menu

menuID	menuType	menuDesc

Dish

dishID	dishName	prepTime	ingredient

Food

dishID	menuID

Question 4

OrderID	Order Date	CustID	Cust Name	Cust Address	Product ID	Product Desc	Product Price	Ordered Quantity
1006	10/11/2016	0002	Kelvin Ng	Sg. Chua, Kajang	7	Dining Table	800.00	1
1006	10/11/2016	0002	Kelvin Ng	Sg. Chua, Kajang	5	Single Bed	400.00	2
1006	10/11/2016	0002	Kelvin Ng	Sg. Chua, Kajang	8	Computer Desk	550.00	1
1007	11/11/2016	0006	Adriana	Samudra, Bangi	11	4-Dr Dresser	1200.00	1
1007	11/11/2016	0006	Adriana	Samudra, Bangi	8	Computer Desk	550.00	1

- **First normal form** → Remove multivalued and composite attributes

Order

OrderID	OrderDate	CustID	CustName	Cust Address
1006	10/11/2016	0002	Kelvin Ng	Sg. Chua, Kajang
1007	11/11/2016	0006	Adriana	Samudra, Bangi

OrderDetails

OrderID	ProductID	ProductDesc	ProductPrice	OrderedQuantity
1006	5	Single Bed	400.00	2
1006	7	Dining Table	800.00	1
1006	8	Computer Desk	550.00	1
1007	8	Computer Desk	550.00	1
1007	11	4-Dr Dresser	1200.00	1

- **Second normal form** → Remove partial functional dependency

OrderDetails

OrderID	ProductID	OrderedQuantity
1006	5	2
1006	7	1
1006	8	1
1007	8	1
1007	11	1

Product

ProductID	ProductDesc	ProductPrice
5	Single Bed	400.00
7	Dining Table	800.00
8	Computer Desk	550.00
11	4-Dr Dresser	1200.00

- **Third normal form** → Remove transitive dependencies

Order

OrderID	OrderDate	CustID
1006	10/11/2016	0002
1007	11/11/2016	0006

Customer

CustID	Cust Name	Cust Address
0002	Kelvin Ng	Sg. Chua, Kajang
0006	Adriana	Samudra, Bangi

Question 5

UNIVERSITI PUTRA MALAYSIA
CLASS GRADE REPORT
SEMESTER II 2024/2025

CourseID: CCS5001
Course Title: Database
Instructor Name: Hazlina Hamdan
Instructor Office: C1.19 Block C, FSKTM

StudentID	Student Name	Major	Grade
38214	Raymond Tan	01 – Computer Science	A-
40873	Qaleesya Ammar	02 – Software Engineering	A
51893	Elena Aaron	02 – Software Engineering	B-
67877	Solahuddin Shauki	03 – Computer Science	B
89887	Sharifah Umairah Syed Hassan	04 – Network	A

1. **First Step**

GradeReport

CourseID	Title	Instructor Name	Instructor Office	StudentID	Student Name	MajorID	Major
CCS5001	Database	Hazlina Hamdan	C1.19 Block C, FSKTM	38214	Raymond Tan	01	Computer Science
CCS5001	Database	Hazlina Hamdan	C1.19 Block C, FSKTM	40873	Qaleesya Ammar	02	Software Engineering
CCS5001	Database	Hazlina Hamdan	C1.19 Block C, FSKTM	51893	Elena Aaron	02	Software Engineering
CCS5001	Database	Hazlina Hamdan	C1.19 Block C, FSKTM	67877	Solahuddin Shauki	03	Computer Science
CCS5001	Database	Hazlina Hamdan	C1.19 Block C, FSKTM	89887	Sharifah Umairah Syed Hassan	04	Network

2. **1NF**

Course

CourseID	Title	InstructorName	Instructor Office
CCS5001	Database	Hazlina Hamdan	C1.19 Block C, FSKTM

StudentGrade

CourseID	StudentID	StudentName	MajorID	Major	Grade
CCS5001	38214	Raymond Tan	01	Computer Science	A-
CCS5001	40873	Qaleesya Ammar	02	Software Engineering	A
CCS5001	51893	Elena Aaron	02	Software Engineering	B-
CCS5001	67877	Solahuddin Shauki	03	Computer Science	B
CCS5001	89887	Sharifah Umairah Syed Hassan	04	Network	A

3. **2NF**

StudentGrade

CourseID	StudentID	Grade
CCS5001	38214	A-

CourseID	StudentID	Grade
CCS5001	40873	A
CCS5001	51893	B-
CCS5001	67877	B
CCS5001	89887	A

Student

StudentID	StudentName	MajorID	Major
38214	Raymond Tan	01	Computer Science
40873	Qaleesya Ammar	02	Software Engineering
51893	Elena Aaron	02	Software Engineering
67877	Solahuddin Shauki	03	Computer Science
89887	Sharifah Umairah Syed Hassan	04	Network

4. **3NF**

Student

StudentID	StudentName	MajorID
38214	Raymond Tan	01
40873	Qaleesya Ammar	02
51893	Elena Aaron	02
67877	Solahuddin Shauki	03
89887	Sharifah Umairah Syed Hassan	04

Major

MajorID	Major
01	Computer Science
02	Software Engineering
02	Software Engineering
03	Computer Science
04	Network

Instructor

InstructorName	Instructor Office
Hazlina Hamdan	C1.19 Block C, FSKTM

Complete:

Course

CourseID	Title	InstructorName
CCS5001	Database	Hazlina Hamdan

Instructor

InstructorName	Instructor Office
Hazlina Hamdan	C1.19 Block C, FSKTM

StudentGrade

CourseID	StudentID	Grade
CCS5001	38214	A-
CCS5001	40873	A
CCS5001	51893	B-
CCS5001	67877	B
CCS5001	89887	A

Student

StudentID	StudentName	MajorID
38214	Raymond Tan	01
40873	Qaleesya Ammar	02
51893	Elena Aaron	02
67877	Solahuddin Shauki	03
89887	Sharifah Umairah Syed Hassan	04

Major

MajorID	Major
01	Computer Science
02	Software Engineering
02	Software Engineering
03	Computer Science
04	Network